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DUHEM AND CONTINUITY IN THE HISTORY OF SCIENCE

Roger ARIEW and Peter BARKER

Pierre Duhem is recognized universally for founding the modern historical study of medieval science. But one of his most important claims, that medieval science is continuous with modern science, has been disputed from the moment he announced it. The debate over the historical and historiographical status of Duhem's thesis continues to the present day (1). We believe the thesis is most compelling as a contingent claim about history of science : continuity just happens to be the case ; it could have been otherwise. Understood in this way, the continuity of medieval and modern science was rejected by most leading scholars in the history and philosophy of science from the 1920s to the present. Antonio Favaro, the editor of *Le Opere di Galileo Galilei*, rejected continuity almost immediately after the death of Duhem (2). The next generation of historians of science, led by Alexandre Koyré and Anneliese Maier (3), rejected the continuity thesis during the 1940s and 1950s, followed by Thomas Kuhn and other historians and philosophers of science in the 1960s

(1) See, for example, Joseph AGASSI, "Towards an Historiography of Science" *History and Theory*, Beiheft 2 (1963).

(2) Antonio FAVARO, "Galileo Galilei e i Doctores Parisienses", *Rencondita della Accademia dei Lincei* 27 (1918) : 3-14 and "Léonard de Vinci a-t-il exercé une influence sur Galilée et son école ?" *Scientia* 20 (1916) : 257-265.

(3) For example, Alexandre KOYRÉ, *Études Galiléennes* (Paris : Hermann, 1939) — *Galileo Studies*, trans. J. Mepham (New Jersey : Humanities Press, 1978) — and the collection of Anneliese Maier's writings translated into English as *On the Threshold of Exact Science*, ed. and trans. S. D. Sargent (Philadelphia : University of Pennsylvania Press, 1982).

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and 1970s (4). The negative appraisal of Duhem's thesis continues to the present, for example in Peter Dear's *Mersenne and the Learning of the Schools*, published in 1988 (5). In appraising the present state of the debate we consider each of these figures in turn. We conclude that the criticisms of Duhem by many of these figures rest upon misapprehensions of his work, and to that extent fail as criticisms of the continuity thesis he stated. In a suitably broadened form, we argue, the continuity thesis deserves to be returned to the active agenda of historians of science.

I

For most of the nineteenth century, scholars treated "medieval science" as an oxymoron. Since nothing from the middle ages was worthy of the name "science", no history of medieval science could be written. This view included even William Whewell, a founding figure in the new genre of history of science : "The two great periods of School Philosophy ... were that of the Greeks and that of the Middle Ages ; — the period of the first waking of science, and that of its mid-day slumber" (6). Modern science appeared during the seventeenth century with the simultaneous rejection of both the "first waking" — Greek science — and the "mid-day slumber", called medieval mysticism. Whewell gave a typical treatment in the following passage from *The History of the Inductive Sciences*. The chapter is entitled "Of the Mysticism of the Middle Ages" :

> a new and peculiar element was introduced into the Greek philosophy which ... tinged a large portion of the speculations of succeeding ages. We may speak of this peculiar element as *Mysticism* ... Thus, instead of referring the events of the external world to space and time, to sensible connexion and causation, men attempted to reduce such occurrences under spiritual and supersensual relations and dependencies ; they referred them to superior intelligences, to theological

(4) Thomas KUHN, *Structure of Scientific Revolutions* (Chicago : The University of Chicago Press, 1962) ; Imre Lakatos, *Philosophical Papers, Volume 1* (Cambridge : Cambridge University Press, 1978).

(5) Peter DEAR, *Mersenne and the Learning of the Schools* (Ithaca : Cornell University Press, 1988).

(6) William WHEWELL, *History of the Inductive Sciences*, third ed. (London, 1857) Book I, Introduction.

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> conditions, to past and future events in the moral world, to states of mind and feelings, to the creatures of an imaginary mythology or demonology. And thus their physical Science became Magic, their Astronomy became Astrology, the study of the Composition of bodies became Alchemy, Mathematics became the contemplation of the Spiritual Relations of number and figure, and Philosophy became Theosophy ... This tendency materially affected both men's speculations and their labours in the pursuit of knowledge ... And by calling into a prominent place astrology, alchemy, and magic, it long occupied most of the real observers of the material world. In this manner it delayed and impeded the progress of true science ; for we shall see reason to believe that human knowledge lost more by the perversion of men's minds and the misdirection of their efforts, than it gained by any increase of zeal arising from the peculiar hopes and objects of the mystics (7).

Whewell's work is typical of the intellectual context in which Pierre Duhem wrote *L'évolution de la mécanique*. Published in 1903, this work dismissed the middle ages as scientifically sterile. Similarly, Duhem's history of chemical combination, *Le mixte et la combinaison chimique*, completed in 1900, jumped from Aristotle's concept of *mixtio* to modern concepts. It was only in 1904, while writing *Les origines de la statique*, that Duhem came across an unusual reference in a 16th-century book to a then unknown medieval thinker, Jordanus de Nemore. His pursuit of this reference, and the research to which it led, is acknowledged on all sides to have created the history of medieval science for the twentieth century (8).

Where his earlier histories had been silent, *Les origines de la statique* contained a number of chapters on medieval thought. One chapter treated Jordanus de Nemore. Another treated his followers. A third argued their influence on Leonardo da Vinci. As Duhem later told us, he discovered that :

> Le Moyen-Age chrétien avait connu des écrits de Statique composés par les Grecs ; certains de ces écrits lui étaient parvenus directement, et d'autres par l'intermédiaire de commentaires arabes. Mais les Latins qui lisent ces œuvres ne sont nullement les commentateurs serviles

(7) WHEWELL, *ibid.*, Book IV, chap. 3. We should note that the whole chapter is only 2 pages long !

(8) For more on Duhem's early historical work, see Anastasios Brenner, *Duhem : Science, réalité et apparence*, (Paris : Vrin, 1990), pp. 131-46.

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> et dénués de toute invention qu'on s'est plu à nous montrer en eux. Les débris de la pensée grecque, qu'ils ont reçus de Byzance ou de la Science islamique, ne demeurent pas en leur esprit comme un dépôt stérile ; ces reliques suffisent à éveiller leur attention, à féconder leur intelligence ; et dès le XIIIe siècle, peut-être même avant ce temps, l'École de Jordanus ouvre aux mécaniciens des voies que l'Antiquité n'avait pas connues (9).

While writing the second volume of *Les origines de la statique* in 1905-6, Duhem greatly extended his historical scope. As expected he covered 17th-century statics, including Galileo, Stevin, and Roberval. But he also returned to medieval statics, spending four chapters on geostatics. Here Duhem continued the history of statics from the Greeks to the moderns, depicting the development of geostatics from Albert of Saxony, in the 14th century, to Torricelli in the 17th (10).

In showing modern statics to be continuous with medieval statics Duhem had scarcely begun his historical task. From 1906 to 1913 he delved deeply into his favorite guide for the recovery of the past, the scientific notebooks of Leonardo da Vinci. In the three-volume *Études sur Léonard de Vinci : ceux qu'il a lus et ceux qui l'ont lu*, he published a series of essays uncovering da Vinci's medieval sources and their influence on the moderns. The third volume gained a new subtitle, *Les précurseurs parisiens de Galilée*, announcing Duhem's bold new thesis that even the works of Galileo had a medieval heritage :

> Lorsque nous voyons la science d'un Galilée triompher du Péripatétisme buté d'un Cremonini, nous croyons, mal informés de l'histoire de la pensée humaine, que nous assistons à la victoire de la jeune Science moderne sur la Philosophie médiévale, obstinée dans son psittacisme ; en vérité, nous contemplons le triomphe, longuement préparé, de la

(9) Pierre DUHEM, "Notice sur les titres et travaux scientifiques de Pierre Duhem, rédigée par lui-même lors de sa candidature à l'académie des sciences (mai 1913)", in *Mémoires de la société des sciences physiques et naturelles de Bordeaux*, series 7, vol. 1 (1917), p. 160. (Pierre Duhem, "Research on the History of Physical Theories", trans. Roger Ariew and Peter Barker, *Synthese* 83 (1990), pp. 190-91 ; *Synthese* 83, no. 2 and 3 are special issues entitled *Pierre Duhem : Historian and Philosopher of Science*).

(10) DUHEM, "Notice", pp. 160-61. ("Research", pp. 191-92).

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> science qui est née à Paris aux XIVe siècle sur les doctrines d'Aristote et d'Averroès, remises en honneur par la Renaissance italienne ([11]).

Duhem went on to present Galilean dynamics as a continuous development out of medieval dynamics. He recovered the late medieval theory of *impetus*, tracing it from John Philoponus' criticism of Aristotle to its mature statements in the 14th-century works of John Buridan and Nicole Oresme :

> Le rôle que l'*impetus* joue, en cette Dynamique de Buridan, c'est très exactement celui que Galilée attribuera à l'*impeto* ou *momento*, Descartes à la *quantité de mouvement*, Leibniz enfin à la *force vive* ; si exacte est cette correspondance que pour exposer, en ses *Leçons académiques*, la Dynamique de Galilée, Torricelli reprendra souvent les raisonnements et presque les paroles de Buridan ([12]).

Duhem then sketched the extension of impetus theory from terrestrial dynamics to the motions of the heavens and earth :

> à la terre, [Nicholas Oresme] attribue un *impetus* naturel semblable à celui que Buridan attribue aux orbes célestes ; pour rendre compte de la chute verticale des graves, il admet que l'on doit composer cet *impetus* par lequel le mobile tourne autour de la terre avec l'*impetus* engendré par la pesanteur. Le principe qu'il formule nettement, Copernic se bornera à l'indiquer d'une manière obscure et Giordano Bruno à le répéter ; Galilée usera de la Géométrie pour en tirer les conséquences, mais sans corriger la forme erronée de la loi d'inertie qui s'y trouve impliquée ([13]).

Duhem went on to describe late medieval contributions to the science of weights, hydrostatics, geology, what was later called the law of fall, and analytic geometry.

Duhem's essay on Leonardo da Vinci concludes with a speculation about the mechanism transmitting medieval ideas to modern science. Since the works of Buridan and Oresme had remained almost entirely in manuscript, Duhem suggested that Albert of Saxony, whose works were printed and reprinted during the 16th century, was the likely link to Galileo. Duhem's key to understanding the transmission of medieval science was Galileo's use of the phrase *Doctores Parisienses*,

(11) DUHEM, "Notice", pp. 162. ("Research", pp. 193).
(12) DUHEM, "Notice", pp. 163-64. ("Research", pp. 194).
(13) DUHEM, "Notice", pp. 166. ("Research", pp. 196).

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a conventional label covering Buridan and Oresme. He proposed that specific 16th-century editions of Albert of Saxony formed the link. Based on evidence including references to certain unusual doctrines and the particular order of the questions, Duhem concluded that Galileo had consulted George Lokert's compilation of Albert of Saxony, Themo Judaeus, and others, and the works of the Dominican Domingo de Soto ([14]).

In the three years before his death in 1916 Duhem wrote *Le système du monde*. The ten volumes were published at irregular intervals over the four decades from 1913 to 1959. These books present an enormous amount of information about medieval astronomy, astrology, tidal theory, and geostatics, again presenting many sources for the first time in our period. They also trace developments in doctrines associated with such concepts as infinity, place, time, void, and the plurality of worlds. Duhem died before completing the project, so we do not have his final thoughts on the continuity thesis.

II

Antonio Favaro was an early and prominent critic of the continuity thesis. As the editor of *Le Opere di Galileo Galilei* his negative appraisal carried special weight. Duhem had constructed a path from the fourteenth century to modern science using some manuscripts of Galileo's. Favaro dismissed these manuscripts as mere juvenilia, and not representative of Galileo's own thoughts. Indeed, he refused to include one of the three manuscripts in his edition of Galileo's works. He did publish the other two, but used the title *Juvenilia* for one of them. Favaro's rejection of continuity influenced the next two generations of historians of science. It is therefore worth pausing to point out that recent scholarship has vindicated Duhem, both in general and in detail.

The work of A. C. Crombie, Adriano Carugo, and William Wallace has shown that the manuscripts dismissed as juvenilia by

(14) Pierre DUHEM, *Études sur Léonard de Vinci*, vol. III (Paris : Hermann, 1913), pp. 582-83.

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Favaro do not date from Galileo's youth ([15]). Wallace has argued cogently that they represent Galileo's notes for his own lectures at the University of Pisa. Crombie and Carugo give them an even later date. These manuscripts contain many 14th-century doctrines, supporting Duhem's original argument for continuity ([16]). One of the manuscripts is a commentary on late medieval interpretations of the *Posterior Analytics*, Aristotle's most influential work on scientific method. The appearance of this material among Galileo's papers suggests that a thesis about the continuity of scientific methodology should be added to Duhem's original thesis about the content of science.

The actual mechanism of transmission has also become clearer from the works of Crombie, Carugo, and especially Wallace. As already noted, Duhem proposed that Albert of Saxony's writings, made popular by George Lokert and transmitted by Domingo de Soto, carried the doctrines of Buridan and Oresme into the sixteenth century. This proposal is substantially correct, although recent research has uncovered other intermediaries. The Jesuits form a bridge between early 16th-century scholastics and Galileo. Of particular importance is Domingo de Soto's brilliant student Franciscus Toletus, who taught natural philosophy at the *Collegio Romano* after he became a Jesuit. As Wallace has shown in great detail, Galileo's manuscripts were partly copied from lecture notes and published works by *Collegio Romano* professors ([17]).

Some of Duhem's claims for Galileo's knowledge of medieval doctrines require revision in the light of recent scholarship. Duhem relied upon references to particular doctrines, and the characteristic order of certain questions, in making his case for continuity. Wallace has shown that this material can also be found in the Jesuit notes

(15) William A. Wallace, *Galileo's Early Notebooks : The Physical Questions. A Translation from the Latin, with Historical and Paleographical Commentary* (Notre Dame : Notre Dame University Press, 1977) ; *Prelude to Galileo. Essays on Medieval and Sixteenth-Century Sources of Galileo's Thought* (Dordrecht : Reidel, 1981). A. C. Crombie and A. Carugo, "The Jesuits and Galileo's Ideas of Science and Nature", *Annali dell'Istituto e Museo di Storia della Scienza di Firenze*, 8 fasc. 2 (1983). S. Drake "Galileo's Pre-Paduan Writings : Years, Sources, Motivations", *Studies in History and Philosophy of Science, 17* (1986) : 429-88.

(16) Cf. Wallace, *Galileo and His Sources.*

(17) William A. Wallace, *Galileo and His Sources.* See also Wallace, "Duhem and Koyré on Domingo de Soto", *Synthese* 83 (1990) : 239-60.

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and texts Galileo was copying. The strongest claim that can be made is, therefore, that Galileo could have acquired this material from essentially contemporary Jesuit sources ; he may not have read Albert of Saxony directly ([18]).

III

Duhem's historical work was almost completely dismissed by the most influential historians of science in the next generation. Anneliese Maier (1905-1971) and Alexandre Koyré (1892-1964) both acknowledged Duhem's studies as the foundation for twentieth century work on medieval science, but subjected those studies to scathing criticism. Similar dismissals of Duhem occur in the work of their students and those influenced by them.

Maier and Koyré's antagonism towards Duhem is the more remarkable because of the many similarities between their work. They trace many of the same figures and themes through similar historical periods. All three employ the pre-eminently French method of *explication de texte*. All favor long quotations from primary sources in their original languages. All three authors conquered mountains of primary sources. Although Koyré wrote the *Études galiléennes* in Cairo ([19]), without access to the extensive primary sources at the disposal of Duhem in Paris, or Maier in Rome, while they wrote the histories for which they are best remembered, any deprivation suffered by Koyré was more than compensated during other periods at Paris and Princeton. Given these similarities of research agenda and techniques, what accounts for the surprisingly vehement criticisms of Duhem by Koyré and Maier ?

Two themes divide the historical writings of Koyré and Maier from Duhem. Both authors present histories of science in which metaphysics plays a primary role in explaining scientific change. Second, the role of metaphysics in science underpins a historiography of early modern science that gives central place to the concept of

(18) W. Wallace, "Galileo Galilei and the *Doctores Parisienses*", in R. E. Butts and J. C. Pitt (eds.), *New Perspectives on Galileo*, (Dordrecht : Reidel, 1978), pp. 87-138.

(19) Charles C. Gillispie, "Alexandre Koyré", in C. C. Gillespie, ed., *Dictionary of Scientific Biography*, vol. 7 (New York : Scribners, 1970-80), pp. 482-90.

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revolution. Medieval thought and early modern science are judged to be different in kind as well as in content. To the extent that Maier and Koyré project these concerns into Duhem's work, they fail to make contact with Duhem as an historical figure or with his contribution to the history and philosophy of science. Let us consider these themes in reverse order.

At the very beginning of his 1949 attack "Le vide et l'espace infini au XIVe siècle", Koyré quotes a passage from Duhem that was to become infamous :

> S'il nous fallait assigner une date à la naissance de la science moderne, nous choisirions sans doute cette date de 1277, où l'évêque de Paris proclama solennellement qu'il pouvaient exister plusieurs mondes et que l'ensemble des sphères célestes pouvait, sans contradiction, être animé d'un mouvement rectiligne. (Duhem, *Études sur Léonard de Vinci*, II, p. 411). [If we are obliged to assign a date to the birth of modern science, we would undoubtedly choose 1277, when the bishop of Paris solemnly proclaimed that a multiplicity of worlds could exist, and that the system of celestial spheres could, without contradiction, be endowed with straight line motion] (20).

Koyré derides the two theses from the 1277 proclamation paraphrased here, calling them two absurdities. He also damns Duhem for failing to recognize the context, which, following Gilson, Koyré takes to be theological. But his greatest scorn is reserved for the date Duhem gives as the birth of modern science, 1277 (21).

Koyré notes in passing that Duhem gives another date elsewhere for the birth of modern science (corresponding to Buridan's impetus theory), but dismisses that date as "aussi fausse que la première" (22).

(20) Cited here as reprinted in Koyré, *Études de l'histoire de la pensée philosophique*. (Paris : Gallimard, 1961) pp. 37-92. Note the distance Duhem places between himself and an actual date : "If we were obliged to assign a date ...". When Duhem returns to the image, in *Le système du monde* VII, p. 4, he even attributes the image to a friend and colleague of his, Albert Dufourcq — Pierre Duhem, *Medieval Cosmology*, ed. and trans. Roger Ariew (Chicago : The University of Chicago Press, 1985), p. 511.

(21) KOYRÉ, "Le vide et l'espace", pp. 37-38.

(22) KOYRÉ, "L'espace et le vide", p. 37, note 3. The reference is to *Études sur Léonard de Vinci*, III, pp. ix-x, repeated in *Le Systéme du Monde*, VIII, where Duhem claims that Buridan's mechanics is an antecedent of Galileo's (p. 200) and that Buridan's impetus theory stands in a similar relation to the inertia principle in seventeenth century physics (p. 338).

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Summarizing the views of Anneliese Maier, Steven Sargent also selects this later date as the date Duhem got wrong : "In short, [Duhem] announced that the scientific revolution of the seventeenth century had actually taken place in the fourteenth century ..." (23).

The fact that Duhem — who prized consistency and was quick to condemn its violation in others (e.g. Maxwell) — spoke of the birth of modern science in connection with several different dates might suggest to a disinterested reader that his language was not literal, but figurative. Duhem employed the trope of nativity to indicate the relative importance of several causes among a multiplicity operative in the generation of modern science (24). Both Koyré and Maier read Duhem as dating the boundary between modern science and a distinct predecessor, surely an odd view to attribute to the creator of the continuity thesis.

Koyré does not object to the denial of continuity implicit in his reading of Duhem, but only to its historical location. For Koyré modern science is distinct and discontinuous from earlier thought. He misreads Duhem because he expects to find such a discontinuity, but cannot accept its location in medieval Paris. For Koyré, this discontinuity appears three centuries after Bishop Tempier, with the abolition of the celestial spheres, and the works of Galileo and Descartes that laid the foundation of modern science (25). He reads Duhem's phrase about "the birth of modern science" literally because for Koyré there must be such a moment.

Maier also rejects one of Duhem's dates for the origin of modern science, a date which she insists ought to mark a break with the past (26). She rejects Duhem's claim that modern science can be dated

(23) Steven D. Sargent, Introduction to *On the Threshold of Exact Science : Selected Writings of Anneliese Maier on late medieval natural philosophy* (Philadelphia : University of Pennsylvania Press, 1982), p. 12.

(24) For a very fine analysis of the tension between complexity and continuity of history for Duhem, a tension that natural clasdsification ultimately would resolve, see Brenner, *Duhem*, pp. 165-66.

(25) Koyré, "Le vide et l'espace", p. 92 ; see also *Études Galiléennes*, Pts. II & III.

(26) "Dr. Maier rejected [Duhem's] contention that the beginning of the scientific revolution occurred in the fourteenth century with such developments as impetus theory ...". E. Grant, "Annaleise Maier (1905-71)" *Archive for the History of the Exact Sciences, 24* (1974) : 144. For a different perspective on the continuity of theories of motion from the medieval to early modern period, see Steven Menn, "Descartes and Some Predecessors on the Divine Conservation of Motion", *Synthese* 83 (1990) : 215-238.

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from Buridan's impetus theory on the grounds that fourteenth century science is distinct from both what preceded and what followed it. For her, scholastic mechanics is, in Sargent's phrase "an independent but transitional stage of development between Aristotelianism and classical physics" ([27]). The separation marks a difference in worldview, and is further indicated by her phrase "on the threshold of exact science" ([28]). For Maier, late medieval natural philosophy is on the threshold, distinct from, but not part of the edifice of modern science ([29]).

Both Koyré and Maier employ the same pattern of argument to support their revolutionary reading of the origins of modern science. They hold that metaphysics provides a needed foundation for the content of empirical science. They claim that early modern metaphysics is discontinuous with medieval metaphysics. Hence, they claim, early modern science is not continuous with medieval thought.

For Koyré, the introduction of Platonic metaphysics — the mathematization of nature — marks a sharp break with the Aristotelian middle ages ([30]). For Maier, it is in the fourteenth century that "The attempt is made for the first time to find principles that permit a direct, individual, and empirical perception and understanding of nature, independent of all authority" ([31]). This new epistemology of

(27) Sargent, Introduction, p. 10. A similar tripartite scheme is suggested by Koyré, *Galileo Studies*, pt. I, p. 3.

(28) Maier, *Die Vorläufer Galileis im XIV Jahrhundert* (Rome : Edizioni di storia e letteratura, 1949).

(29) Duhem himself also uses the image of threshold ; talking about Gregory of Rimini's theory of the infinite in *Le système du monde* VII, pp. 135-36, he states : "Si Grégoire de Rimini a entrevu et souhaité une Mathématique des multitudes et des grandeurs transfinies, il ne faudrait pas, cependant, exagérer son œuvre jusqu'a en faire l'avant-coureur de la théorie que M. Georges Cantor devait construire ... ; désireux de pénétrer en l'Arithmétique des multitudes infinies, il n'a pas su, cependant, depasser le seuil de cette science". (Duhem, *Medieval Cosmology*, p. 113).

(30) Koyré makes this point by opposing Aristotle to Archimedes. It is clear, however, that for Koyré Archimedes is a Platonist (*Galileo Studies*, p. 38), and that Platonism is the general metaphysics underlying modern science. See especially *Galileo Studies*, pp. 204-05, where Koyré says of Galileo, "His science is Plato's revenge" (p. 208). Maurice Clavelin's analysis of Koyré's critique of Duhem makes essentially the same points, though it is somewhat more favorable to Koyré's thesis of the mathematization of nature ; see Clavelin, "Le débat Koyré-Duhem, hier et aujourd'hui", *History and Technology* 4 (1987) : 13-37.

(31) Maier, (tr.) Sargent, *On the Threshold of Exact Science*, pp. 146-7.

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nature separates fourteenth century natural philosophers from their Aristotelian predecessors. Although it prepares the way for the Renaissance, the new method of knowing nature is also distinct from the modern one, as it does not yet incorporate the modern ideal of the mathematical representation of nature. Quite apart from the historical merits of these claims, they fail as criticisms of Duhem unless he also accepts the thesis that metaphysics forms the foundation of empirical science at every period in its history. This is very far from Duhem's view. Duhem consistently excludes metaphysics from science ; however, for Duhem this exclusion is a norm for which we should strive, rather than an historical fact. To understand Duhem's attitude to these episodes we must see them against the background of his general views on metaphysics, consistently maintained throughout his scientific, philosophical, and historical writings.

Many episodes in the history of science display metaphysical elements ; Cartesian physics in the seventeenth century and the Lorentz electron theory in the nineteenth are examples. There is also a long tradition of metaphysical debate in philosophy and theology, which Duhem neither ignores nor discounts. Indeed his attitude towards metaphysics in science is largely the result of his evaluation of the history of metaphysics. He sees the history of metaphysics as the history of warring factions, none of which ever win decisive victories. Unlike science, where all workers finally agree on the value of, say, the principle of energy conservation, metaphysics never reaches consensus. The continual indecisive strife of metaphysics also makes it impossible to claim that there is progress in metaphysics. To permit or encourage the metaphysical elements that have sometimes appeared during the history of science would be to risk destroying scientific consensus and eliminating the possibility of scientific progress.

Such an outcome would be doubly tragic, for it would eliminate our best hope for establishing the correct metaphysics. Science clearly has metaphysical implications ([32]). Science — as an historical activity

(32) "Il serait déraisonnable de travailler au progrès de la théorie physique si cette théorie n'etait pas le reflet, de plus en plus net et de plus en plus précis, d'une métaphysique ; la croyance en un ordre transcendant à la physique est la seule raison d'être de la théorie physique". Duhem, "La valeur de la théorie physi-

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— also displays consensus and progress. The mistake is in thinking that the metaphysical implications of any particular period show us the nature of ultimate reality. Only at the hypothetical endpoint of scientific progress will the empirically grounded categories of our scientific theories correspond to the metaphysically ultimate categories. It is not the nature of any particular scientific theory that is evidence for this ; rather it is the history of science as a whole, and in particular the impression that (the most important parts of) science can be understood as progressing towards such a unified picture. The category scheme of endpoint science, which corresponds to the metaphysical categories of ultimate reality, Duhem calls the "natural classification" ([33]).

As a consequence, any scientific theory that claims to describe the fundamental nature of reality is objectionable. Cartesian atomism, in the seventeenth century, and the classical theory of the electron developed by Lorentz and Poincaré, in the nineteenth, are such theories. These theories may well contribute to the progress of science, but their contribution will be new empirical laws, not "fundamental entities". Furthermore, any new scientific theory must contribute to the already established direction of progress in science. Theories that conspicuously fail to do this are to be rejected. An example is Maxwell's theory of electricity and magnetism, which is discontinuous with previous work. In an extended analysis of contemporary theories of electricity and magnetism, Duhem endorses Helmholtz's theory, and rejects Maxwell's, on the grounds that the former but not the latter can be seen as a continuation of the foundational theories of Poisson and Ampère ([34]). The case of

que", in *La théorie physique, son objet et sa structure* (Vrin : Paris, 1981), p. 509 (*Aim and Structure of Physical Theory*, trans. P. Wiener (Princeton : Princeton University Press, 1954), p. 335). See also Duhem, "Physique de croyant", in *La théorie physique*, p. 460 (*Aim and Structure of Physical Theory*, p. 303).

(33) DUHEM, *La théorie physique*, Pt I, chapter 2 ; "Physique de croyant", part 7.

(34) Duhem also criticizes Maxwell for logical inconsistencies, incoherencies, vagueness, lack of system, and overdependence on models. For Duhem's criticism of Maxwell, see P. Duhem, *Les théories électriques de J. Clerk Maxwell : étude historique et critique* (Paris : Hermann, 1902), and Roger Ariew and Peter Barker, "Duhem on Maxwell : a case study in the interrelations of history of science and philosophy of science", in A. Fine & P. Machamer (eds.) *PSA 1986 : Vol. 1* pp. 145-156 (East Lansing, Michigan : Philosophy of Science Association, 1986).

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electromagnetism presents an unusually clear example of Duhem's reasoning on scientific continuity, but his other work makes the theme equally plain. As a normative principle, we should exclude from science those theories that do not form part of the historical progression. Theories that introduce metaphysical novelties, which are themselves incompatible with the tradition, should be the first to go.

As historical implications of consensus and progress in science, we should expect of find that only the non-metaphysical parts of theories like Cartesian physics and Lorentz electron theory become permanent parts of science. A second class of theories, for example the thermodynamics based on potential functions developed by Duhem, takes no metaphysical risks. Any new results achieved by such theories may therefore be confidently accepted as contributing to the progress of science towards its endpoint (providing, of course these theories are acceptable on other grounds). Duhem believed that his researches provided additional evidence for the presence of these patterns in the history of science. Theories that introduce metaphysical novelties ought not to contribute to the progress of science and historically they did not.

Let us return now to Koyré's and Maier's objections to Duhem. Both claim that modern science is discontinuous and distinct from medieval thought. Both support this claim by asserting the influence of metaphysics on the content of science, and insisting on discontinuities in the metaphysics underlying the successors to medieval science. For Koyré, the Platonism of modern science replaces Aristotelianism. For Maier, fourteenth century dynamics rests on a new epistemology of nature, separating it from medieval thought. Duhem can readily admit these changes. They are quite consistent with his view of the history of metaphysics as ceaseless indecisive conflict. What he would deny is that these changes had any lasting consequences for science.

It might be objected here that the influence of metaphysics in a given episode of scientific history can be documented by historical research. But this is to miss the point of Duhem's analysis of the long-term movement of science. If one believes that metaphysical changes have long term implications for science, how can scientific opinion be unanimous over the long term ? This is a question that

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Koyré and Maier cannot answer, despite their belief in scientific progress ([35]).

The argument from scientific consensus excludes any lasting role for metaphysics in science, and hence for scientific revolutions understood as the replacement of one metaphysics by another. But Duhem's account also excludes revolutions understood as discontinuities in the content of science, at the level of empirical law or theory. Only laws or theories that contribute to the established historical progression towards the "natural classification" will become permanent parts of science. Their continuity with this progression, established since the time of Aristotle and the Greeks, prohibits the introduction of concepts or laws that contradict their immediate predecessors. The tradition may transform over time, as the Aristotelian concept of *potentia* transformed into the richer concept of a potential function in nineteenth century physics, but such change is gradual and continuous ([36]). For Duhem, scientific revolutions form no part of the history of science, a position he defends at the level of philosophical argument and through historical evidence.

Both Maier and Koyré criticize Duhem for mis-dating the revolution that gave birth to modern science. But in Duhem's account of science there can be no revolutions, so he can hardly be blamed for getting their dates wrong. Both authors' insistence that Duhem *must* be employing their own preferred historical categories, despite clear systematic implications to the contrary in Duhem's work as a whole, shows the extent to which the criticisms of Koyré and Maier are levelled at their own misreadings and not at Duhem. Whatever the merits of the thesis that metaphysics decisively influences science, Koyré and Maier's criticisms of Duhem seem to be instances of the same principle at work in the writing of history. Their metaphysical commitment to revolutionary accounts of science obliges them to read Duhem as a bad revolutionary account, rather than what it is, an account in which the influence

(35) Koyré, *Études galiléennes*, p. 1 ; Maier, in Sargent, *On the Threshold of Exact Science*, pp.63, 75, 170.

(36) The claim to continuity is best understood as the claim that the concept developed through the actions of continuously operating causes, among which the historical influence of the Aristotelian concept plays a major — but not exclusive — role. See Duhem, "Physique de croyant", pp. 462-72 (*Aim and Structure of Physical Theory*, pp. 305-11).

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of metaphysics is specifically excluded, and revolutionary change is denied.

IV

Koyré and Maier became sources themselves for revolutionary accounts of science in Kuhn and his successors. In his celebrated book of 1962, *Structure of Scientific Revolutions*, Kuhn generalized Koyré's account of the revolution that founded modern science, portraying the whole history of science as a succession of tranquil eras under the hegemony of a "paradigm", punctuated by abrupt discontinuities (37). Although his book incorporated history in a philosophical account of science more thoroughly than anyone since Duhem, the account was achieved only at the expense of considerably distorting certain historical episodes (38). Much of the attraction of his work lay in its explicit incorporation of metaphysical elements absent from the logical empiricist philosophy of science then dominant in the English-speaking world. The addition of metaphysics to the contents of science led to the outcome Duhem had warned about : the destruction of consensus and the impossibility of scientific progress. The perpetual metaphysical conflict foreseen by Duhem reappeared in Kuhn as relativism — the impossibility of choosing between the metaphysical alternatives represented by different paradigms. So acute was this difficulty that Kuhn was able to say nothing about the progress of science as an historical whole, and very little about the mechanism by which the consensus that existed under one paradigm was recreated under a new one. The latter problem created a whole new genre of sociological accounts motivated by the perception that philosophical models were impotent to explain scientific consensus (39). However, Kuhn's work had the salutary

(37) Clavelin also notes the influence of Koyré on Kuhn and Hanson ; see Clavelin, "Le débat Koyré-Duhem", p. 25.

(38) There is a considerable literature on the philosophical and historical limits of Kuhn's account ; see, for example, B. R. Gholson and P. Barker, "Kuhn, Lakatos and Laudan : Applications to the history of physics and pschology". *American Psychologist*, 40 (1985) : 755-59 ; P. Barker and B. R. Goldstein, "The Role of Comets in the Copernican Revolution", *Studies in History and Philosophy of Science*, 19 (1988) : 299-319.

(39) B. Barnes, *T. S. Kuhn and Social Science* (New York : Columbia University Press, 1982). D. Bloor, *Knowledge and Social Imagery*. London : Routledge, 1976. B. Latour, *Science in Action*. Cambridge, Mass. : Harvard University Press, 1987.

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effect of widening the terms of debate about the nature of science from the categories of "theory" and "evidence" to which the logical empiricists had limited themselved. Paradigms supplied answers to questions about ontology, epistemology, and methodology, in addition to inspiring theories in an ongoing dialogue with experiment and observation. The location of methodology as an element within a paradigm, and hence an element that changed when the paradigm changed, also undermined the logical empiricist vision of scientific method as a universal atemporal set of rules.

Recent critics of Duhem have been influenced by the social studies of science that grew out of Kuhn's work, and by the philosophical implication that there is no universal scientific method. A notable example is Peter Dear's *Mersenne and the Learning of the Schools.* Dear's implicit rejection of the continuity thesis is more subtle than Koyré's or Maier's, and seems to be the articulation of a widely-shared view.

Dear distinguishes between two continuity theses, continuity of theory and continuity of method. He dismisses continuity of theory, claiming that although elements of medieval science did find "their way to Galileo, his enterprise differed sufficiently to render them of only secondary value in understanding the development of his mechanics" ([40]). Thus, lack of continuity is due to changes in the context of science, and most importantly in its goals or its methods, even though some theories remain outwardly unchanged. Dear accepts continuity of method, but dismisses the appeal to method as an historical oversimplification :

> The strongest [of the objections to the methodological varieties of the continuity thesis] is that "science" differs radically from "applied scientific method". As Paolo Rossi has written, one cannot explain the Scientific Revolution in terms of the development or application of correct scientific method, because such a method — taken as a regulative, determinate procedure generating objective knowledge of nature — does not exist ; science is not that simple ([41]).

Even if the existence of Dear's "regulative determinate procedure generating objective knowledge of nature" could be definitively

(40) DEAR, *Mersenne*, p. 232. (See note 5).
(41) DEAR, *Mersenne*, p. 234-35.

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settled (42), the significance of this conclusion for historians of science would still need to be considered. Dear speculates that Wallace and other proponents of methodological continuity are motivated by an image of science as an accumulation of true results, achieved by applying such a method. Whether or not this is a correct reading of such work, the case for continuity may be evaluated quite apart from its supporters' motivations. On the other hand, in his attack on anachronism, Dear correctly argues that the truth or falsity of a theory, and correctness or incorrectness of a method, cannot legitimately form the basis for the explanations given by historians :

> People do not believe things *because* they are true. Consequently, quite apart from Rossi's surely correct observation that there is no "scientific method" or set of rules for producing true knowledge, we many say that even if there were, it could never, even in principle, assist in producing accounts or explanations of Galileo's work in particular or the Scientific Revolution in general (43).

Ultimately, Dear asserts that "true continuity would reside in a continuity of natural philosophical agendas, presupposing a continuity not only of specific ideas but also of goals" (44). This is an overly restricted view of "true continuity". Taking the continuity of large-scale philosophical agendas as the criterion of "true continuity", we would have to conclude that there is no continuity between Galileo and Descartes, or between Mersenne and Descartes, or between early and late Galileo, for example. Because Dear's notion of continuity is so restrictive, it fails as a criticism of Duhem. But, more importantly, Dear has not excluded many other forms of continuity that historians may feel obliged to acknowledge, not in general, but in particular cases.

Discontinuities of context, method, or goals, do not entail that the continuity thesis is false ; in particular historical cases, continuity may not require identical contexts, methods, or goals. As we have argued, Duhem himself could readily have conceded all three

(42) On the philosophical sources of the illusion that there is universal, atemporal scientific method : P. Barker, "The reflexivity problem in the psychology of science". In B. Gholson *et al., Psychology of Science : Contributions to Metascience*, pp. 92-114. (Cambridge : Cambridge University Press, 1989).

(43) DEAR, *Mersenne*, p. 235.

(44) DEAR, *Mersenne*, p. 238.

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discontinuities, yet maintained his continuity thesis in its original form. He presented many historical examples of methods and goals that had changed discontinuously. He understood that Peripatetics, atomists, Cartesians, Newtonians, British electromagnetic theorists, German electromagnetic theorists, and supporters of the Lorentz electron theory disagreed about both methods and goals. He insisted, however, that these differences made no permanent contribution to science. Stripped of these superfluous elements, anything that remained might contribute to the progress of science towards its endpoint, the natural classification. Consequently, in *La théorie physique*, Duhem argued for instrumentalism, or the doctrine that physical theory is classificatory as opposed to explanatory ([45]). The only continuity Duhem ever claimed was that connecting empirical laws ([46]). Physical theory in Duhem's sense need not and should not be wedded to any particular metaphysics or philosophical agenda. Thus, any compelling historical argument for such continuities would do violence to Duhem's fundamental conviction that what was best about science had an historical integrity quite separate from the history of these other factors.

In the very first chapter of *La théorie physique*, Part I, Duhem declared that "Aucun système métaphysique ne suffit à édifier une théorie physique" ([47]). This thesis is the precursor to Dear's point of view, that no scientific method suffices in constructing a physical theory. Both views are generalizations of what is now called the Duhem-Quine thesis ([48]), that observations do not suffice in deter-

(45) P. DUHEM, *La théorie physique*, chap. 1-2. Duhem distinguishes sharply between classification, which occurs in science, and explanation, which he confines to metaphysics. Explanation can proceed only from the one correct ontology. Science cannot make ontological claims, although the categories introduced by science consistute a classification of nature that approximates more and more closely to the ultimate metaphysical categories as science progresses.

(46) Duhem did not acknowledge the existence of entities at the level of Kuhn's paradigms, for example. Nor did he allege continuity at the level of methods, ontology, goals, or agendas. Duhem's denial that changes in ontology, etc., contribute to the progress of science would have made it impossible for him to admit entities like Kuhn's paradigm's or Lakatos's research programs, as we have argued elsewhere. See R. Ariew and P. Barker, "Duhem on Maxwell".

(47) DUHEM, *La théorie physique*, p. 17 (*Aim and Structure of Physical Theory*, p. 16).

(48) For the relations between the Duhem thesis and the Duhem-Quine thesis, see Roger Ariew, "The Duhem Thesis", *British Journal for the Philosophy of Science* 35 (1984) : 313-25.

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mining particular laws. Duhem repeatedly emphasized that observation underdetermines hypotheses, laws, and theories. His favorite example was the observational equivalence of epicycles and eccentrics in ancient astronomy ([49]).

Underdetermination can be extended to include all levels of science. Physical theories underdetermine and are underdetermined by metaphysical theories. The same may be said of theories and methods. Underdetermination also occurs between theories and goals. Thus, Dear's own point of view can be used to argue that a single theory might persist through changes in context, or changes from one philosophical agenda to another. If underdetermination is as general as we suspect, then historians may assume neither continuity nor discontinuity in advance of examining the evidence. In some historical cases, hypotheses, laws, or theories may be continuous despite changes in goals, contexts, or agendas ([50]). We conclude that the issue of continuity or discontinuity can only be resolved by historical investigation, and that any decision will be limited to the case examined ([51]).

Given generalized underdetermination, we need not share Duhem's conviction of the importance of excluding metaphysics from science. But the issue that motivates this exclusion, the nature of scientific consensus, reappears as the question of relativism in studies of science today. As the durability of Duhem's insight here shows, his history and philosophy of science is more rich, and more sophisticated than many now recognize. His detractors have often failed to understand the full richness of his thought, and like Koyré and Maier, attacked

(49) See, for example, Duhem's *Sozein ta phainomena : essai sur la notion de théorie physique* (Vrin : Paris. 1982) — *To Save the Phenomena*, trans. E. Doland and C. Mashler, (Chicago : University of Chicago Press, 1969).

(50) Examples might include the Maxwell-Lorentz theory of electromagnetism, which survived the demise of classical physics and its replacement by relativity theory — E. Zahar, *Einstein's Revolution*, (La Salle, IN : Open Court, 1989) — and the Lorentz electron theory itself, which originated in a Newtonian context and survived into the brief era of the "electromagnetic world-view" — R. McCommach, "H. A. Lorentz and the Electromagnetic View of Nature", *Isis* 61 (1970) : 459-97.

(51) For the kind of historical studies we mean to refer to, see the articles in *Revolution and Continuity : Essays in the History and Philosophy of Early Modern Science*, edited with Introduction by Peter Barker and Roger Ariew (Washington, DC : Catholic University of America Press, 1991).

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phantoms that are really projections of their own historiographies. Even if they were right in claiming that there were fundamental changes in metaphysical doctrine during the late medieval and early modern periods, this does not automatically require the falsity of a thesis of continuity. Moreover, the assumption, too easily accepted in studies of science today, that synchronized conceptual, theoretical or methodological discontinuities are a regular, recurrent feature of science, fails to withstand historical scrutiny and is not forced upon us by any compelling philosophical consideration. The concept of continuity brought forward by Duhem remains an indispensable tool for understanding some, if not all, of the history and present state of science.

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